

DRILLING PROJECT MANAGEMENT

By:



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Content

- Introduction to Project Management & Drilling Project Management
- Modul-1 JDS Project Initiation

Project :

@ PMI

- “ A **temporary endeavor** undertaken to create a **unique** product, service, or result.”

Temporary : A project has a definite beginning and end.

Endeavor : It involves planned work and coordinated activities.

Unique product, service, or result : The outcome is distinct from routine operations.

Each project is temporary, goal driven, and produces a specific deliverable.

What is Project Management?

Project Management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements."



Integration

Connecting thousands of small details into one cohesive whole.



Leadership

Setting the direction and keeping the team on path to the goal.



Risk Mitigation

Anticipating the storm before it arrives at the worksite.



Value Creation

Ensuring every cent spent delivers a result for the company.

Why do we need PM? (The Order from Chaos)

Without Project Management, a drilling project will likely suffer from:

- **Scope Creep:** Work expanding without direction.
- **Budget Overrun:** Costs swelling beyond the AFE.
- **Missed Deadlines:** Wasted time on the rig floor.
- **Chaotic Communication:** Miscoordination between vendors and the team.

Key Message: "PM brings **ORDER** to **CHAOS**. It transitions us from being reactive (putting out fires) to being proactive (preventing the fire from starting)."

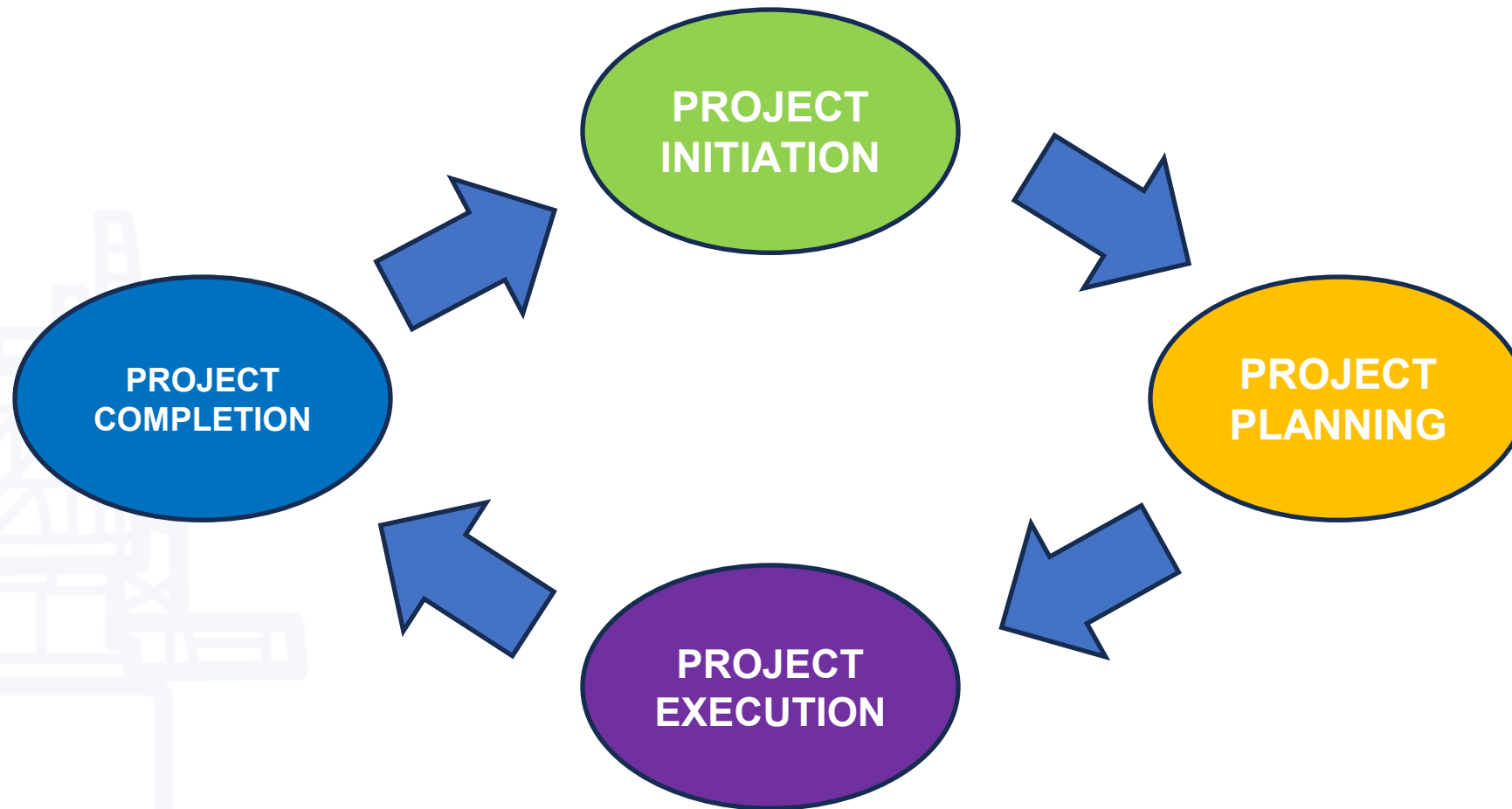
Triple Constraint



- **Scope:** What work is being done? (Depth, logging program, completion).
- **Time:** How long will it take? (Days vs. Depth, Rig release date).
- **Cost:** What is the budget? (AFE, daily spread rate).

A change in one constraint inevitably impacts the other two. Quality is the result of balancing these three

Project Life Cycle





Exercise : Provide Simple Project

Wedding Party

Project Phase	Analogy
Initiation	Decision to marriage, Rough Budget Estimate.
Planning	Venue decision, Catering Services decision, Wedding Organizer, Invitation letter, etc.
Execution & Monitoring	D-Day (Resepsi/Akad Nikah).
Completion	Final Invoicing & Vendor Evaluation

* Imagine if the catering **didn't show up** (Operational Risk) or the venue **wasn't paid off** (AFE Overrun)



Why PM Matters in Drilling?

DPM as "The Art of Integration"

In drilling, the Project Manager is the "glue" that holds dozens of specialized service companies and internal departments together.

- **The Concept:** Drilling involves coordinating 30+ vendors (mud, logging, cementing, etc.) simultaneously.
- **The Key Message:** DPM is about ensuring that Subsurface, Engineering, Procurement, and Operations are perfectly aligned

Managing "Subsurface Uncertainty"

Unlike a construction project where you can see the worksite, drilling happens thousands of feet underground in an invisible environment.

- **The Concept:** We rely on probabilistic models (P10, P50, P90) because we cannot see the target directly.
- **The Key Message:** DPM is not just about following a plan; it is about managing the risks of geological "surprises" like pressure anomalies or hazardous gases

Value Protection (NPT & ILT)

In other industries, a delay might just be an annoyance. In drilling, "Time is Money" is a literal, expensive truth.

- **The Concept:** Every hour of **Non-Productive Time (NPT)** or **Invisible Lost Time (ILT)** can burn thousands of dollars in rig spread rates.
- **The Key Message:** The Project Manager's primary goal is to protect the investment by ensuring every minute on the rig floor is productive and safe

Project Life Cycle - DPM

PROJECT INITIATION

- Project Scope
- Risk Assessment project
- Project Economical

PROJECT PLANNING

- Drilling Basis of Design
- Project timeline
- Detail Budget / AFE
- Risk Register
- Logistic & Supply chain plan
- Procurement Strategy
- Well Control Plan
- Emergency & Contingency Plan

PROJECT EXECUTION & MONITORING

- Drilling Program
- Bridging document
- DWOP (Drilling Well on Paper)
- Bill of Material (BOM)
- Daily Drilling Report (DDR) & NPT/ILT Tracking
- Cost & Contract Monitoring

PROJECT COMPLETION

- End of Well report
- Drilling Post Mortem
- Well Performance Analysis
- After Action Review
- Service Quality Meeting
- Financial Close Out

Application in Drilling

- **Managing Subsurface Uncertainty:** Unlike construction, we cannot see our "worksite" clearly. We manage geological risks using probabilistic models (P10, P50, P90).
- **The Stakeholder Landscape:** Managing relationships with Regulators (SKK Migas), Partners, Service Companies, and Local Communities.
- **The Cost of Inefficiency:** In drilling, "Time is Money" is literal. Every hour of **NPT (Non-Productive Time)** can cost thousands of dollars.
- **Risk-Based Decision Making:** Using *Decision Gates* to ensure the project only proceeds if it remains safe and economically viable

Project Management Team

Role	Main Responsibility
Project Manager (PM)	Document Integrity, <i>Decision Tree</i> , Team Coordination
Subsurface Specialist	G&G Analysis, Coordinate Target, PPFG profiles, Offset Well Data
Drilling Design Engineer	Well Architecture Design, Rig Selection, Trajectory design, dll
Risk & HSE Engineer	Hazard Identification, Risk Matrix, Contingency / emergency Plan, HSE Bridging document
Project Contract & Economist	Cost Estimation (<i>Dry Hole vs Complete</i>), <i>Spread Rate calculation</i> , AFE Cost

“Even when the scope of work is typical, every project is unique.”

Two wells may follow the same drilling program, but the project execution and challenges will never be exactly identical;

- Different well conditions.
- Different reservoir characteristics.
- Different rig or equipment.
- Different environmental and regulatory requirements
- Different stakeholders and timelines
- Different risks and operation challenges.

Organization

- Project Management Institute : PMP®, CAPM®,
- AXELOS / PeopleCert : PRINCE2®

Brake – 10 minutes

By:

PROJECT INITIATION

By:

Project Initiation

Laying the Strategic and Economic Foundation for Advance Drilling Operations and Value Creation

Establish the “Why” and “How” Before the first bit touches the ground

Core Objectives

- **Technical Feasibility:** Validating if the geological target can be safely reached with current technology
- **Economic Viability:** Ensuring the cost of drilling justifies the potential hydrocarbon recovery
- **Strategic Alignment:** Confirming the project fits into the company's long-term energy portfolio



Defining Project Scope

Technical Boundaries

Establishing precise surface and subsurface coordinates, target formations, and depth parameters (TVD/MD) to prevent scope creep during planning

Major Deliverables

Defining the “End State” of the project: Data Sets (Logging/Coring), Well Test result, and the final status (Producer or P&A)

Macro Risk Assessment



Subsurface

Assessing geological uncertainties like pressure anomalies, faulting, and hazardous gas zones



Operational

Evaluating Rig availability, Lead times material, and logistical accessibility to remote sites



Regulatory

Ensuring compliance with SKK Migas permits, environmental licenses, and social impact requirements

Financial Target

25%

Target IRR

Investment Threshold

A Project must exceed the Weighted Average Cost of Capital (WACC). Initiation focuses on high-level NPV modeling and Payback Period Estimation

$$NPV = \sum_{n=0}^N \frac{CF_n}{(1+r)^n}$$

Where:

N = the total number of periods

n = a single period between 0 and N

CF = the cash flow in period n

r = the discount rate

Resource Probability Matrix

Confidence Level	Metric	Est. Reserves (MMBOE)	Cost-Reward Ratio
P90 (Proven)	Conservative	1.2-1.5	Low Risk / High Stability
P50 (Probable)	Base Case	2.8-3.2	Optimal Strategic Target
P10 (Possible)	Upside	5.5-7.0	High Reward / Exploratory

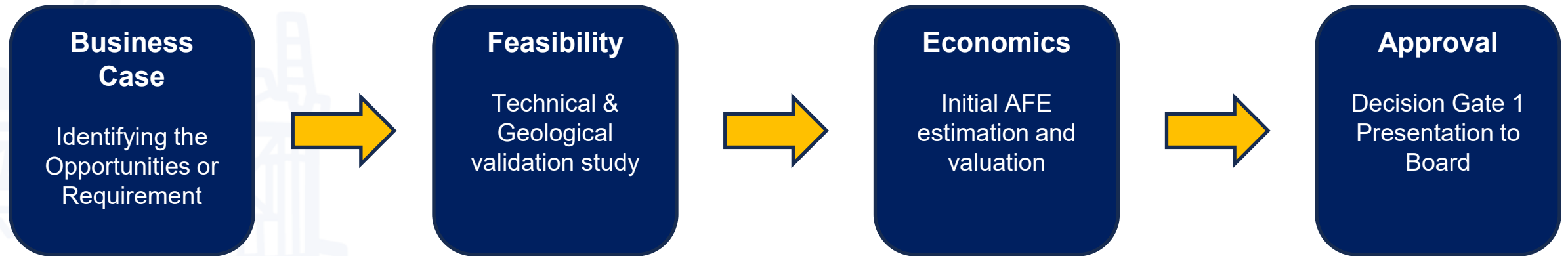
Drilling Proposal

The definitive outcome of Phase-1. It synthesizes all technical and commercial findings into a single business case for executive management.

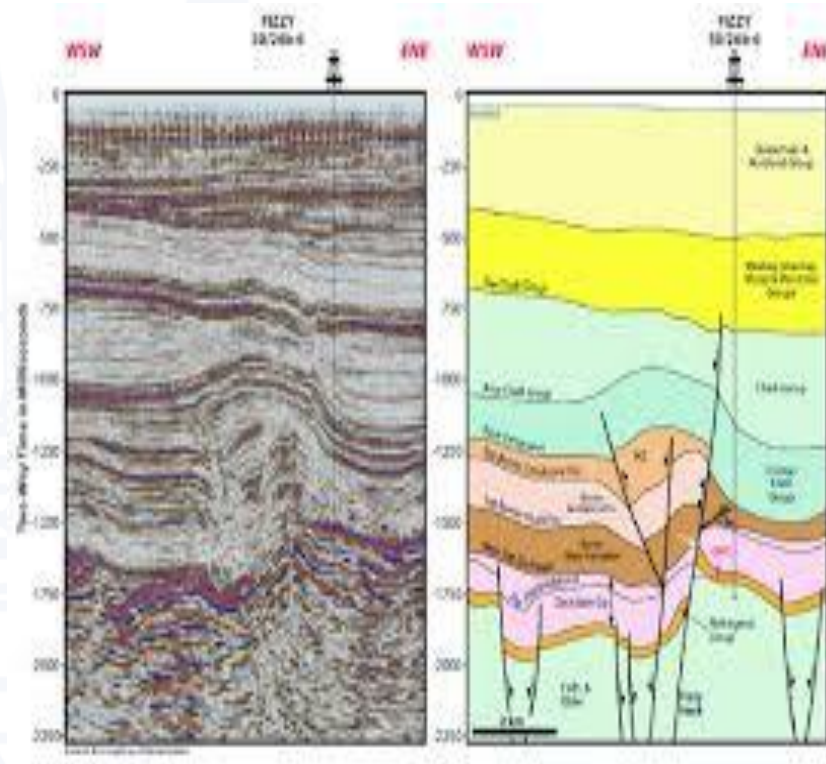
Goal: Secure the initial AFE mandate and move into Phase-2 Planning



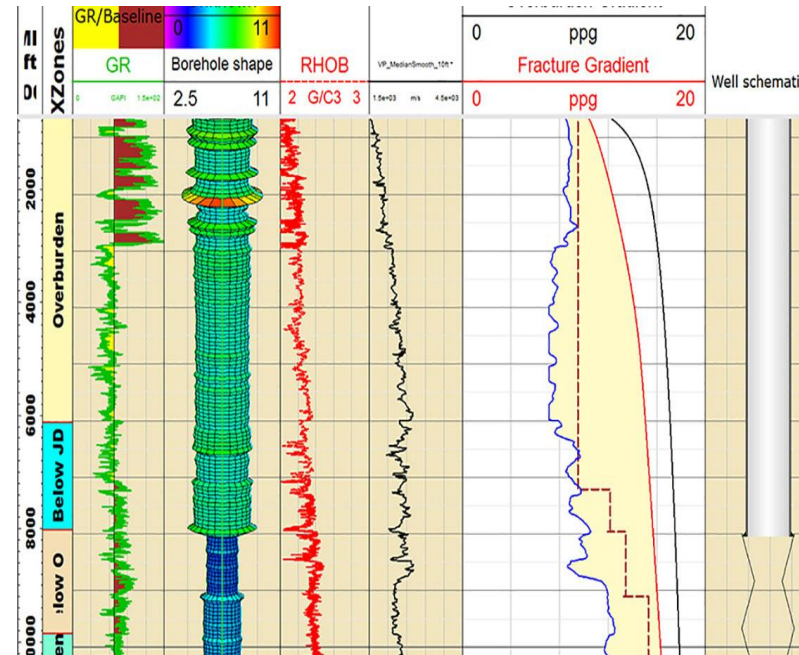
Initiation Workflow



Key Subsurface Inputs



Seismic Interpretation



Pore Pressure Prediction



Offset Well Analysis

Decision Gate 1

The “GO” or “NO-GO” Moment

**Moving from Strategic Concept to
Detailed Engineering**

Q&A

???

PROJECT ASSIGNMENT

By:

Project Assignment

- Phase-1 Drilling Proposal
 - Project Approval
- Phase-2 Basis of Design
 - Peer Review
- Phase-3 Drilling Program
 - DWOP (Drilling Well on Paper)
- Phase-4 N/A

Phase-1 Drilling Proposal

Diskusi Kelompok

Back up

By:

Sesi Pagi

Waktu	Durasi	Aktivitas	Detail Konten
08:00 - 08:15	15'	Opening & Vision	Sambutan Menteri DPM mengenai standar JDS & pengenalan struktur kabinet.
08:15 - 09:00	45'	DPM Fundamentals	Perbedaan Proyek vs Operasi, Triple Constraint, & Lifecycle Pemboran.
09:00 - 09:15	15'	Coffee Break	Istirahat sejenak.
09:15 - 10:00	45'	Phase 1: Initiation	Membedah <i>Drilling Proposal</i> : Justifikasi teknis, risiko makro, dan keekonomian sumur.

Sesi Siang

Waktu	Durasi	Aktivitas	Detail Konten
13:00 - 13:20	20'	PMT Establishment	Workshop: Pembagian 10 posisi kunci dalam kelompok (RACI Matrix).
13:20 - 14:10	50'	Drafting Exercise	Diskusi Kelompok: Menyusun draf poin-poin <i>Drilling Proposal</i> didampingi Mentor.
14:10 - 14:50	40'	Group Pitching	3 Kelompok presentasi draf (7' presentasi + 6' feedback dari Menteri/Wamen).
14:50 - 15:00	10'	Closing & Mandate	Instruksi pengerjaan dokumen final & info jadwal sidang Gate-1.

Diskusi Kelompok

1. Berkumpul sesuai kelompok
2. Diskusi anggota kelompok dan tugas masing-masing anggota
3. Diskusi mengenai draft Drilling Proposal
4. Presentasi, perkenalan kelompok + draft drilling proposal yang akan dibuat